GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

# Similar Triangles; Indirect Measurement

*Independent Set 6.07* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** Are triangles with sides 4, 10, 12 and 6, 15, 20 similar? Show the three ratios.

***No — 6/4 = 15/10 = 1.5, but 20/12 ≈ 1.67***

*Two ratios agree and the third does not.*

**2.** Triangle PQR is similar to triangle XYZ. PQ is 8, QR is 10 and PR is 13, and XY is 20. Find YZ and XZ.

***YZ = 25 and XZ = 32.5***

*The names match PQ with XY: k = 20 ÷ 8 = 2.5; YZ matches QR and XZ matches PR.*

**3.** A triangle has sides 15, 25 and 30. A similar triangle's longest side is 6. Find its other two sides.

***3 and 5***

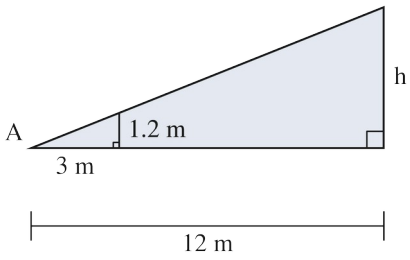
*k = 6 ÷ 30 = 0.2.*

**4.** A 1.5-m child casts a 2-m shadow. At the same moment a tree casts a 14-m shadow. How tall is the tree?

***10.5 m***

*k = 14 ÷ 2 = 7; 1.5 × 7.*

**5.** The two right triangles are similar and share corner A. The small one runs 3 m along the ground from A and stands 1.2 m high. The big one runs 12 m along the ground from A. How high is it?




***4.8 m***

*Both from A: k = 12 ÷ 3 = 4, and 1.2 × 4 = 4.8.*

**6.** Which is similar to a triangle with sides 8, 10 and 14: one with sides 10, 12, 16, or one with sides 12, 15, 21? Show why.

***12, 15, 21 — every side × 1.5***

*10, 12, 16 adds 2 to every side.*